Data models for complex business cases are difficult to develop and maintain. Proper modeling requires expertise in both data modeling and the subject matter. In strongly centralized architectures, especially in big companies, central data teams serve too many domains to secure the required expertise. Their workload is usually too high to allow them to spend sufficient time on resource-intensive tasks of data harmonization and tightly coupled model evolution (we noted these problems previously in chapter 1). Even if we could imagine the cross-functional project of the model development, assuming that the data team is not directly involved in domain operations, then the cognitive load of continuous updating and modifying would strain their capacity.

KEY POINT A data mesh is for companies with complex domains.

The solution is the decentralization of data ownership. You should consider the implementation of a data mesh because domain-focused data ownership is one of its pillars.

DATA DIVERSITY

Harmonization problems often plague companies operating with data of varied formats and from multiple sources (e.g., a combination of structured, unstructured, and semi-structured sources; or as events, files, graphs, relational and NoSQL databases, etc.). Each new data format and source in parallel increases the coupling and chances that a change will occur. Every change in the source or transitional data model may lead to a cascading, more or less catastrophic failure.

The other often-encountered problem is ensuring that the data can be found and accessed. Keeping the metadata repository up-to-date becomes a challenge. The same holds true for the maintenance of granular access rights to hundreds of data assets, especially if data comes with varying levels of protection depending on its origin (e.g., the European Union versus Bolivia).

KEY POINT A data mesh is for companies with diverse data sources.

Suppose the company struggles with consistency, harmonization, and abundant data sources. In that case, a data mesh may be a viable solution